

5Q)Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code)

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/time.h>
int main() {
    int fd[2];
    pid_t pid;
    int data[5] = {10, 20, 30, 40, 50};
    struct timeval start, end;
    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }
    gettimeofday(&start, NULL);
    pid = fork();
    if (pid < 0) {
        perror("fork");
        return 1;
    }
    if (pid > 0) {
        close(fd[0]); // Close read end
        printf("Producer (Parent): Sending data...\n");
        for (int i = 0; i < 5; i++) {
            write(fd[1], &data[i], sizeof(int));
            printf("Produced: %d\n", data[i]);
        }
        close(fd[1]);
        wait(NULL);
        gettimeofday(&end, NULL);
        double time_taken =
            (end.tv_sec - start.tv_sec) * 1000000.0 +
            (end.tv_usec - start.tv_usec);
        printf("\nCommunication time: %.2f microseconds\n",

```

```

        printf("\nCommunication time: %.2f microseconds\n",
               time_taken);
    }
    else {
        close(fd[1]);    // Close write end
        int value;
        printf("\nConsumer (Child): Receiving data...\n");
        while (read(fd[0], &value, sizeof(int)) > 0) {
            printf("Consumed: %d\n", value);
        }
        close(fd[0]);
        exit(0);
    }
    return 0;
}

```

Output)

```

sai@kornypc:~$ ./p5
Producer (Parent): Sending data...
Produced: 10
Produced: 20
Produced: 30
Produced: 40
Produced: 50

Consumer (Child): Receiving data...
Consumed: 10
Consumed: 20
Consumed: 30
Consumed: 40
Consumed: 50

Communication time: 1534.00 microseconds
sai@kornypc:~$ |

```